

Toronto Bike-Share Ridership Prediction Using 2023-2024 Data

Andy Hwang

Department of Computer Science

University of Western Ontario

London, Ontario, Canada

ihwang9@uwo.ca

251217976

Abstract—Bike-share systems (BSS) have become an essential component of urban mobility, yet operators continue to face challenges balancing bikes and docks at individual stations, particularly during peak hours when demand flows unevenly toward city centers. While existing research has primarily focused on system-level demand or relied on data from various cities to develop spatial predictors, most prior work on Toronto’s BSS remains exploratory or focuses on system-wide forecasting rather than station-level demand. Additionally, past studies often rely on simplistic models such as linear regression, lacking the complexity needed to capture variability in ridership.

This study addresses these gaps by developing station-level, hourly demand prediction models for Toronto’s bike-share system using open data. The study evaluates various statistical, machine learning, and deep learning models, examines spatial and temporal features, and incorporates meteorological variables to better understand ridership patterns. The methods include linear regression, polynomial regression, Lasso regression, Ridge regression, Random Forest, XGBoost, and a small neural network. Performance is assessed using time-series cross-validation with RMSE, MAE, and R^2 to guide the selection of effective predictors of local demand and provide insights to support more efficient rebalancing strategies for Toronto’s bike-share system.

Index Terms—bike-share systems, demand prediction, machine learning, deep learning, Toronto, urban mobility

I. INTRODUCTION

Over the past decade, bike-share systems (BSS) have grown increasingly popular in major cities around the world, providing an affordable and environmentally friendly transportation option. In Toronto, bike-share ridership has increased by more than 1 million trips each year, for the past 4 years in a row. Additionally, there are now 1000+ active bike stations in 2024, compared to the 278 that operated in 2018.

However, any user of BSSs will tell you that, especially during peak hours, it can be challenging to either find bikes or find open spots to park one’s bike, resulting in a longer trip than necessary or the need to manually relocate bikes. Toronto’s open data portal on bike-share ridership found that more than 80% of trips at peak hours tend to go in the same direction, towards the downtown core. This is a very common problem for cities, as commuters traditionally have to manually reallocate bikes, while also using point systems to incentivize users to bring bikes to stations that require more bikes, thus balancing the load. Nonetheless, this is still one of the most significant issues faced by BSSs across many

cities, which is why much of the current literature surrounding BSSs, focuses on forecasting the demand and flows of BSSs, so operators can better match capacity with demand.

This study will aim to apply key learnings regarding spatial-temporal features for BSS demand predictions from other papers, to the context of Toronto’s BSS, as a station-level hourly basis. Various statistical, machine learning, and deep learning models will be tested as a ridership baseline.

The rest of this paper is structured as follows: Section 2 reviews related literature. Section 3 details the methodology and dataset preprocessing. Section 4 presents the experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

II. RELATED WORK

A. Bike-Share Demand Modeling Across Cities

Many existing studies focus on the BSS of cities like New York, Seoul, Montreal, or Washington D.C., incorporating models that attempt to predict demand using both spatial and temporal features, such as temperature, wind speed, and date/time-based features. They’ve also discovered ridership far depletes near its center-suburban split, as distance from home of interest (POI) and the center of the city increases. These findings are true especially for commuter-based ridership. However, cross-city results are not always easily applicable as bike-share users’ behaviours can change depending on the location. There is a lack of station-level forecasting. While some papers make different systems reactive quickly differently to the same temperatures. This highlights the need for city-specific models, tailored so local data rather than findings generalizing across systems.

B. Prior Work on Toronto’s Bike-Share System

Several papers attempt to tackle the challenge of bike-share prediction using data from Toronto. These studies examine variables like weather, time of day, seasonality, and other key regional trends, such as seasonal variation, weather effects, and hourly changes. While these papers do highlight important features to be included in models, they often do not train models that predict station-level demand for Toronto, and instead, focus on spatial-temporal ridership, rather than station-level

demand. This makes the results from these studies unsuitable for application to real operations.

C. Station-Specific Forecasting with Graph Convolutional Networks to predict flow on a per-station basis.

Station-specific forecasting with Graph Convolutional Networks to predict flow on a per-station basis. It takes into account station-specific features like station-level history, network effects, latitude and longitude, hourly ridership in previous days, and other variables. These models can sometimes be highly specific to the city, although to incorporate some of these features for each station.

D. Lacking a Testing Toronto-Specific Models

A large gap in the literature is that there are very few papers which test various Toronto-specific prediction models to handle the unique differences of Toronto as compared to other major cities. Second, there is a lack of station-level forecasting. While some papers make predictive models, it's on a system-wide basis. Lastly, most research is using old data. Between the start of 2022 and the end of 2024, there has been an increase of more than 255 stations in Toronto. This is more than 25% of the total number of stations, reflecting Toronto's BSS's increased ridership of Lazbon. Thus, this study will look to provide, Toronto-specific, station-hourly forecasting, using only the most recent data from 2023-2024.

III. PROPOSED METHOD

A. Dataset Description

The dataset used in this project was sourced from Toronto's open data portal and has minute-to-minute trip data for all trips from 2016-01 to 2024-09. The columns in this dataset are as follows: Start Station Id, End Station Id, Start Station Name, End Station Name, Start Time, End Time, Bike Id, User Type, Trip Duration

The General Bikeshare Feed Specification (GBFS) also provides station-level data. The columns for this data are: station_id, name, lat, lon, capacity, is_renting, is_charging_station, nearby_station.

B. Weather Data

Weather data was obtained from the Government of Canada's historical climate data portal. This data included many variables including temperature, precipitation, humidity, and wind speed. In order to accurately reflect the regional climate, we chose the VVCX weather station, which provides hourly data for Toronto's city center.

C. Data Cleaning and Preprocessing

1) *Cleaning*: To clean the trip data, we removed all trips that had invalid or null start or end stations. We also cleaned up station metadata to append latitude, longitude and station capacity. We decided to remove trips that were less than 1 minute in length to the same station as those are likely not representative of real demand and are more likely to be idle rides in which a bike has not been moved. Additional cleaning steps included verifying timestamps and accounting for the changes required for Daylight Savings Time.

D. Feature Engineering

Based on our findings during our exploratory data analysis, we selected several features to test the effectiveness of their to predicting demand:

Temporal Features: To capture the changes from day to day, I one-hot-encoded the days of the week, in order to represent potential differences such as weekdays versus weekends. I also encoded the hour as this was an additional feature. To capture hourly-based trends, I applied cyclical feature encoding using sine and cosine transformations so that hours of the day, as well as a "is_peak_am" and "is_peak_pm" to help capture the two different peaks that occur during a workday.

Weather Features: Based on existing literature, we chose to include temperature, humidity, wind speed, and precipitation as continuous variables. We also one-hot-encoded weather in relation as holidays, as this could point to many riding trips.

Station Features: Station features like observations are included on a per-station basis was station type and proximity to the center. Rather than include latitude and longitude as values, we decided to use distance from city center as a proxy. In hopes to roughly estimate the capital features. While this is not the same citywide, I'll be looking to use clustering approaches programming interfaces (APIs) to locate stations within distance to POIs.

E. Feature Selection:

We are still in the midst of experimenting with different features.

F. Train/Test Split and Validation

As we are working with a time-series, we did not apply random splitting. Instead, we used the typical approach which involves time-series cross-validation. This will also improve our model selection with

G. Model Selection and Training

We are still in the midst of experimenting with different models and comparing results. On a high-level, the models we are testing are listed as below. We will focus more specifically in the final paper.

Evaluation Metrics:

- RMSE
- MAE
- R²

REFERENCES

- [1] THIS WILL BE INCLUDED IN THE FINAL REPORT. SOURCES ARE PREPARED, JUST NOT INCLUDED YET.